



RV COLLEGE OF ENGINEERING

**Design and Prototype Development of Variable Pitch
Propeller**

AEROSPACE PROPULSION

AS343AI



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In partial fulfilment for the award of degree of

Bachelor of Engineering

Aerospace Engineering

2024-2025

RV COLLEGE OF ENGINEERING®, BENGALURU-59

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DEPARTMENT OF AEROSPACE ENGINEERING



CERTIFICATE

Certified that the Project Based Lab component of Aerospace Propulsion AS343AI course titled '**Design and Prototype Development of Variable Pitch Propeller**' is carried out by **A Uday Kiran (1RV23AS004), Rutuj Dugad Manoj (1RV23AS014), Tejas L (1RV23AS061), Shanthosh K V (1RV23AS053), Yashwanth R (1RV23AS064)** who are Bonafide students of R.V. College of Engineering, Bengaluru, in partial fulfillment of the award of degree of **Bachelor of Engineering in Aerospace Engineering** of Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated into the project-based laboratory report deposited in the departmental library. The report has been approved as it satisfies the academic requirements in respect of Aerospace Propulsion Lab prescribed by the institution for the said degree.

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DECLARATION

We, A Uday Kiran (1RV23AS004), Rutuj Dugad Manoj (1RV23AS014), Tejas L (1RV23AS061), Shanthosh K V (1RV23AS053), Yashwanth R (1RV23AS064), students of Fourth semester B.E., Department of Aerospace Engineering, R.V. College of Engineering, Bengaluru, hereby declare that the Project Based Lab of Aerospace Propulsion AS343AI course titled '**Design and Prototype Development of Variable Pitch Propeller**' has been carried out by us and submitted in partial fulfilment for the award of degree of Bachelor of Engineering in Aerospace Engineering during the year 2024-25.

Further, we declare that the content of the report has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of the institute and we will be one of the authors of the same.

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Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.



Abstract

This project presents the design, simulation, fabrication, and experimental validation of a variable pitch propeller system aimed at improving thrust generation across a range of operational conditions. Unlike fixed-pitch systems, variable pitch propellers enable real-time adjustment of blade pitch angles to optimize aerodynamic efficiency and propulsion performance. The study begins with a detailed aerodynamic analysis using Blade Element and Momentum Theory, followed by parametric modeling in SolidWorks and simulation through CFD using Flow Simulation tools. A total of 16 pitch configurations ranging from 0° to 30° were evaluated to understand their influence on thrust.

The prototype was fabricated using 3D printing techniques in PLA material, and experimentally tested using a brushless DC motor, load cell, and a tachometer-based thrust measurement setup. Experimental data showed a strong correlation with simulation results, with the maximum thrust of 3.00 N observed at a pitch angle of 28° , closely matching simulated predictions. The results confirmed that increased pitch angles lead to enhanced thrust until aerodynamic stall effects appear. The study highlights the feasibility of manually tunable propeller blades and lays the groundwork for future development of servo-actuated dynamic pitch systems for UAVs and other aerospace platforms.

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Chapter 1

Introduction

Variable pitch propellers represent one of the most significant technological advancements in propulsion systems, fundamentally transforming how aircraft and marine vessels optimize their performance across varying operational conditions. Unlike fixed-pitch propellers that maintain constant blade angles, variable pitch propellers enable real-time adjustment of blade pitch angles, allowing for optimal thrust generation and efficiency across different flight phases or operating speeds.

The concept of variable pitch propulsion emerged from the inherent limitations of fixed-pitch systems, which can only achieve optimal efficiency at a single design point. In aviation, aircraft experience dramatically different operating conditions during takeoff, climb, cruise, and landing phases, each requiring different thrust and efficiency characteristics. Similarly, marine vessels encounter varying sea conditions, speeds, and loading scenarios that benefit from pitch adjustment capabilities.

The development of variable pitch propeller technology has been driven by the continuous pursuit of improved fuel efficiency, enhanced performance, and operational flexibility. Modern variable pitch systems incorporate sophisticated control mechanisms, advanced materials, and precise manufacturing techniques to achieve remarkable levels of performance optimization. These systems have become standard equipment in commercial aviation, military aircraft, and high-performance marine applications.

This report provides a comprehensive examination of variable pitch propeller technology, exploring the theoretical foundations, design principles, control systems, and practical applications that define modern propulsion systems. The analysis encompasses both aerospace and marine applications, highlighting the unique challenges and solutions associated with each domain.

1.1 Propeller Aerodynamics and Hydrodynamics

The operation of variable pitch propellers is governed by fundamental principles of fluid dynamics, whether operating in air for aircraft or water for marine vessels. Understanding these principles is essential for comprehending how pitch variation affects propeller performance and overall system efficiency.

1.1.1 Pitch Angle Effects

The pitch angle β fundamentally determines the propeller's thrust and torque characteristics. Increasing pitch angle generally increases the theoretical advance per revolution but also increases the angle of attack, potentially leading to stall conditions. The relationship between pitch angle and performance is complex and depends on operating conditions.

The geometric pitch P at radius r is related to the pitch angle by:

$$P = 2\pi r \tan(\beta) \quad (1.1)$$

The advance ratio J is a dimensionless parameter that characterizes the propeller's operating condition:

$$J = \frac{V}{nD} \quad (1.2)$$

where V is the advance velocity.

1.2 Variable Pitch Mechanisms

Variable pitch propellers employ various mechanical systems to adjust blade pitch angles during operation. The choice of mechanism depends on application requirements, including response time, precision, reliability, and power consumption.

1.2.1 Hydraulic Systems

Hydraulic pitch control systems are widely used in aviation applications due to their high power-to-weight ratio and rapid response capabilities. These systems typically employ engine oil or dedicated hydraulic fluid to actuate pitch change mechanisms.

The basic hydraulic system consists of a governor unit that senses propeller speed and controls oil flow to a pitch change mechanism within the propeller hub. The pitch change mechanism converts hydraulic pressure into mechanical motion that rotates the propeller blades about their

longitudinal axes.

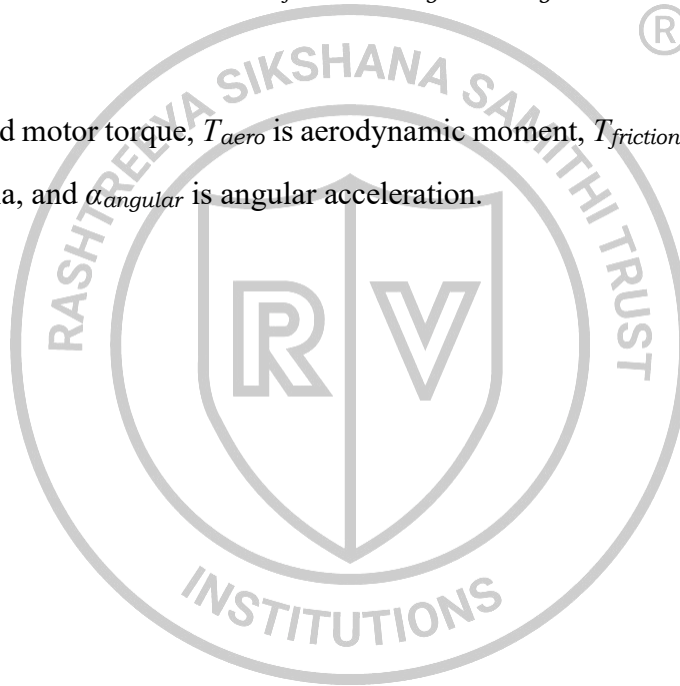
1.2.2 Electric Systems

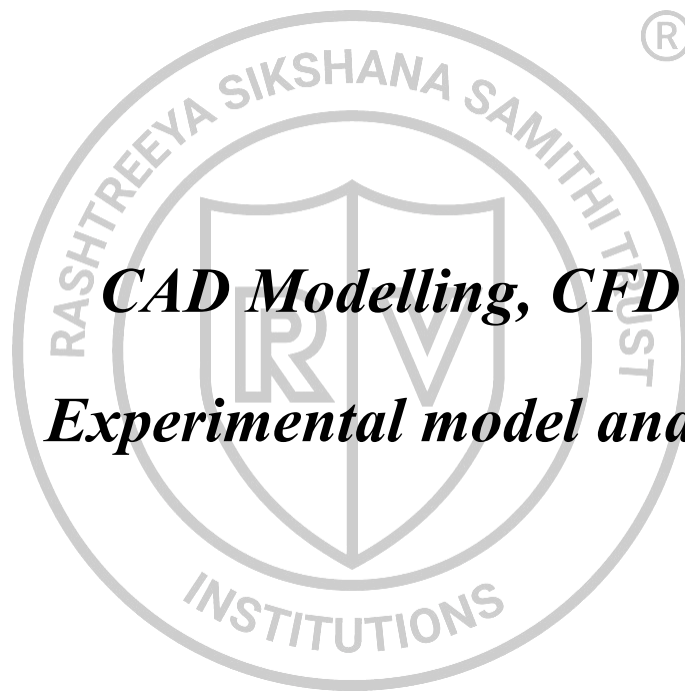
Electric pitch control systems offer precise control and are increasingly used in modern applications. These systems employ electric motors, typically stepper motors or servo motors, to drive pitch change mechanisms through gear reduction systems.

The torque requirements for electric pitch control systems must overcome both aerodynamic moments and system friction:

$$T_{motor} = T_{aero} + T_{friction} + J_{system}\alpha_{angular} \quad (1.3)$$

where T_{motor} is required motor torque, T_{aero} is aerodynamic moment, $T_{friction}$ is friction torque, J_{system} is system inertia, and $\alpha_{angular}$ is angular acceleration.





Chapter 2

CAD Modelling, CFD Analysis, Experimental model and Analysis

2.1 Blade Design and Geometry

The design of variable pitch propeller blades requires careful consideration of aerodynamic efficiency, structural integrity, and manufacturing constraints. Unlike fixed-pitch blades, variable pitch blades must maintain acceptable performance across a wide range of pitch angles.

2.1.1 Twist Distribution

Blade twist distribution must be optimized for variable pitch operation, considering that the effective twist changes as pitch angle varies. The total twist at any radial station is the sum of built-in geometric twist and pitch angle adjustment.

The optimal twist distribution for maximum efficiency can be approximated by:

$$\beta(r) = \beta_{root} - \frac{2}{\pi} \arctan \left(\frac{r_{hub}}{r} \right) \quad (2.1)$$

where β_{root} is root pitch angle and r_{hub} is hub radius.

2.1.2 Geometric Parameters

Number of Blades: The number of propeller blades affects thrust distribution, vibration characteristics, and manufacturing complexity. Fewer blades generally provide higher efficiency but may result in increased vibration and noise. More blades offer smoother operation and better thrust distribution but at the cost of increased complexity and potential efficiency penalties.

Rotor Diameter: The propeller diameter is one of the most critical design parameters, directly affecting the propeller's thrust-producing capability and efficiency. Larger diameters generally provide better efficiency due to lower disk loading, but are constrained by ground clearance, structural considerations, and installation requirements.

Hub Diameter: The hub diameter defines the central portion of the propeller and affects the blade root design, structural attachment, and flow characteristics near the hub. The hub-to-tip ratio influences the effective working area of the propeller and affects both performance and

structural design.

2.1.3 Operating Conditions

Rotation Speed (RPM): The propeller rotational speed determines the blade tip speed and affects both aerodynamic performance and structural loads. Higher rotational speeds can provide more thrust from a given diameter but may encounter compressibility effects and increased noise at high tip speeds.

Ship Speed (Forward Velocity): This parameter defines the advance velocity of the propeller through the fluid medium. It is crucial for determining the advance ratio, which characterizes the propeller's operating condition and affects the angle of attack distribution along the blade.

Required Thrust: The design thrust requirement determines the propeller's loading and influences blade geometry optimization. This parameter is typically derived from the vehicle's drag characteristics and performance requirements.

Fluid Density: The density of the operating medium (air or water) affects thrust production, power requirements, and Reynolds number. This parameter varies with altitude, temperature, and atmospheric conditions for aircraft, or with water temperature and salinity for marine applications.

2.1.4 Computational Grid Parameters

Radial Panels: This parameter defines the number of computational divisions along the blade span from hub to tip. More radial panels provide higher resolution in the spanwise direction, allowing for more accurate representation of blade geometry and flow characteristics. However, increased panel count also increases computational time and complexity.

Chordwise Panels: This parameter specifies the number of computational divisions along the blade chord from leading edge to trailing edge. Higher chordwise panel counts improve the accuracy of pressure distribution calculations and airfoil representation, particularly important for capturing leading edge and trailing edge effects.

2.1.5 Blade Geometry Parameters

r/R (Radial Position): This dimensionless parameter represents the radial position along the blade span, where r is the local radius and R is the propeller tip radius. Values range from the hub ratio (typically 0.15-0.25) to 1.0 at the tip. This parameter defines the stations where blade properties are specified.

c/D (Chord-to-Diameter Ratio): This parameter defines the local blade chord length as a fraction of the propeller diameter. The chord distribution affects blade loading, structural characteristics, and manufacturing requirements. Optimal chord distributions balance aerodynamic efficiency with structural and manufacturing constraints.

C_d (Drag Coefficient): The local drag coefficient represents the airfoil section's drag characteristics at each radial station. This parameter is typically derived from airfoil data and affects the propeller's power requirements and efficiency calculations.

t/D (Thickness-to-Diameter Ratio): This parameter defines the maximum blade thickness at each radial station as a fraction of the propeller diameter. Blade thickness distribution affects structural strength, manufacturing requirements, and aerodynamic performance.

Skew: Blade skew refers to the circumferential offset of blade sections at different radial positions. Skew is used to reduce noise, vibration, and cavitation by distributing blade loading more evenly in time. Forward skew moves outer blade sections ahead of inner sections in the direction of rotation.

Xs/D (Skew Distribution): This parameter quantifies the skew distribution along the blade span. The skew distribution affects the propeller's acoustic signature, vibration characteristics, and hydrodynamic performance.

2.1.6 Parametric Design using OpenProp

The design process begins with processing input parameters:

Specifications		Blade Design Values					
Number of blades:	3	r/R	c/D	C_d	t_0/D	Skew	X_s/D
Rotation speed (RPM):	6000	0.2	0.056	0.008	0.0078	0	0
Rotor diameter (m):	0.42	0.3	0.064	0.008	0.0066	0	0
Required thrust (N):	20	0.4	0.072	0.008	0.0057	0	0
Ship speed (m/s):	15	0.5	0.078	0.008	0.0046	0	0
Hub diameter (m):	0.065	0.6	0.082	0.008	0.0038	0	0
Fluid density (kg/m ³):	1.225	0.7	0.082	0.008	0.0030	0	0
# radial panels:	20	0.8	0.077	0.008	0.0022	0	0
# chordwise panels:	20	0.9	0.064	0.008	0.0014	0	0
		0.95	0.049	0.008	0.0010	0	0
		1	0.001	0.008	0	0	0

Figure 2.1: OpenProp Input Parameters

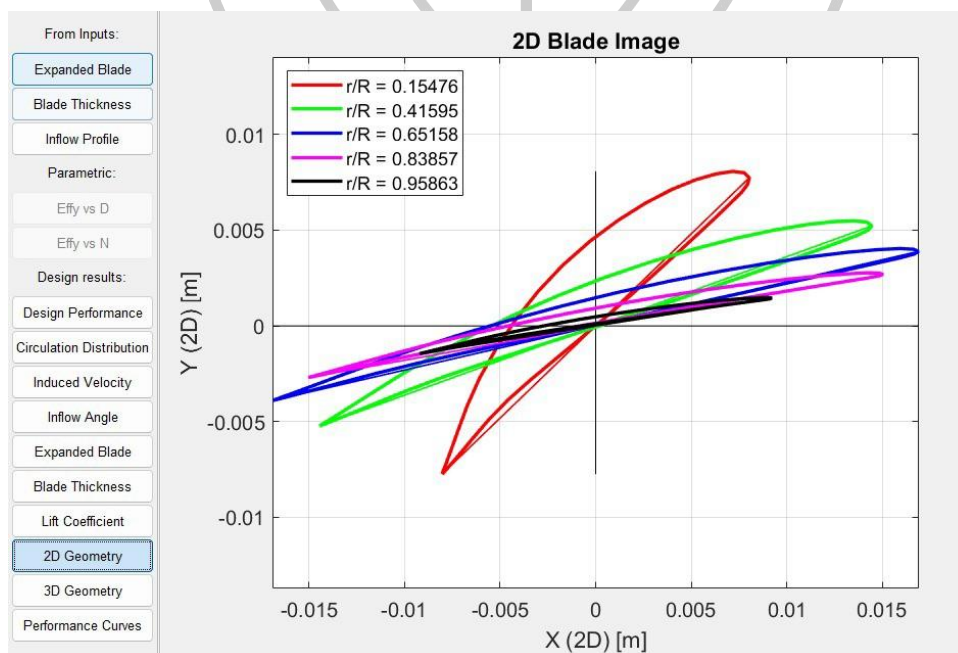


Figure 2.2: Propeller Profiles

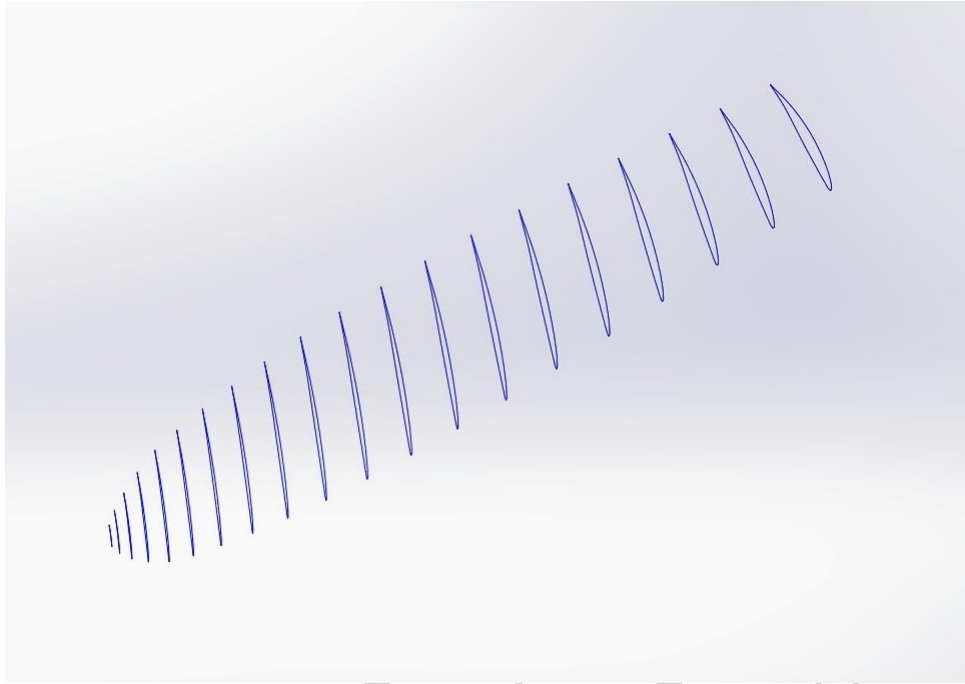


Figure 2.3: Propeller Profiles in SolidWorks

2.2 CAD Design

SolidWorks was utilized to design the propeller hub mechanism. Multiple part files were created and assembled into a comprehensive assembly model.

Key components designed include the hub, coupler, blades, bolts, and supporting structures.

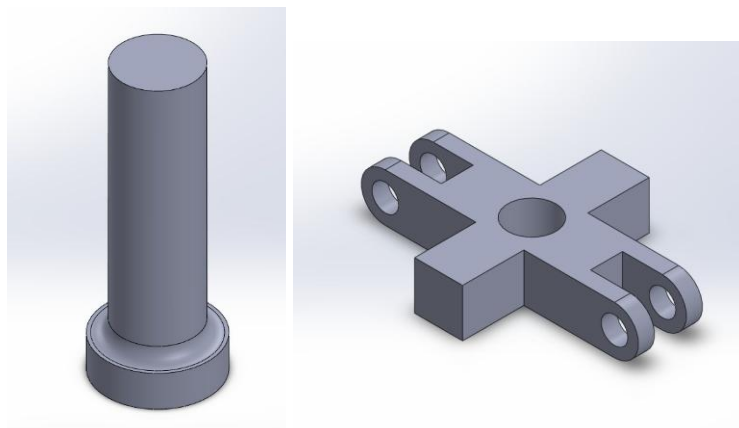


Figure 2.4: Bolt design and pitch controller

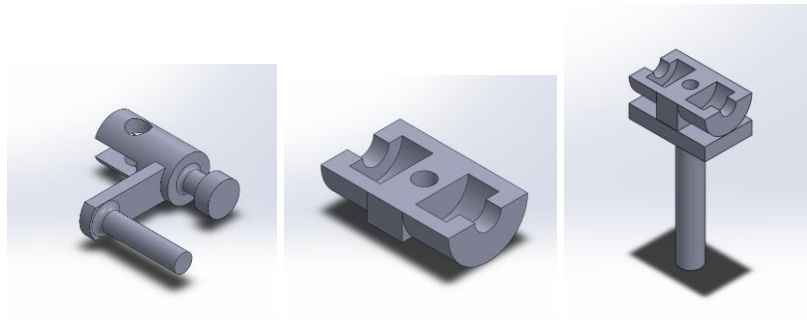


Figure 2.5: Coupler design and hub design

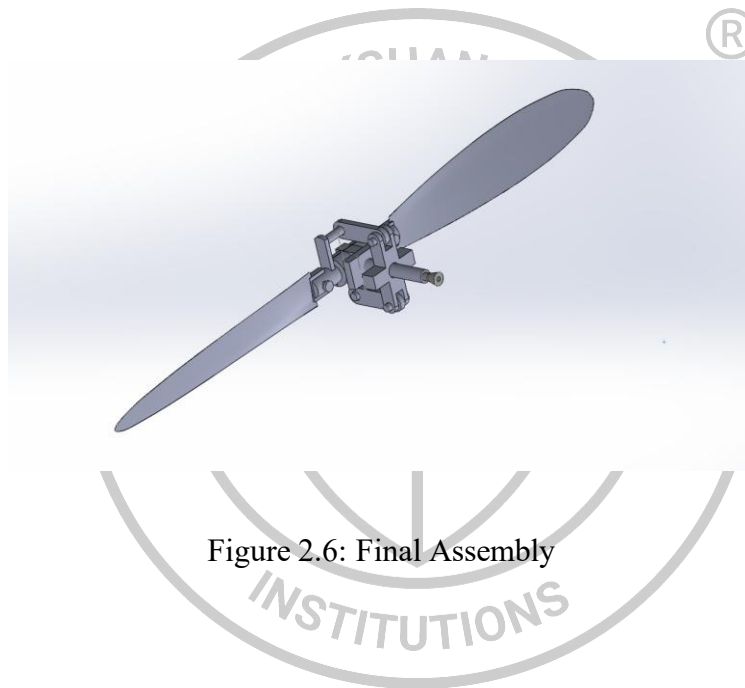


Figure 2.6: Final Assembly

2.3 CFD Simulations

For computational fluid dynamics (CFD) analysis, SolidWorks Flow Simulation was employed. Appropriate boundary conditions were defined, including an environmental pressure setup. All simulations were performed at an operating speed of 3000 RPM to evaluate aerodynamic performance.

Below are the related simulation and model images:

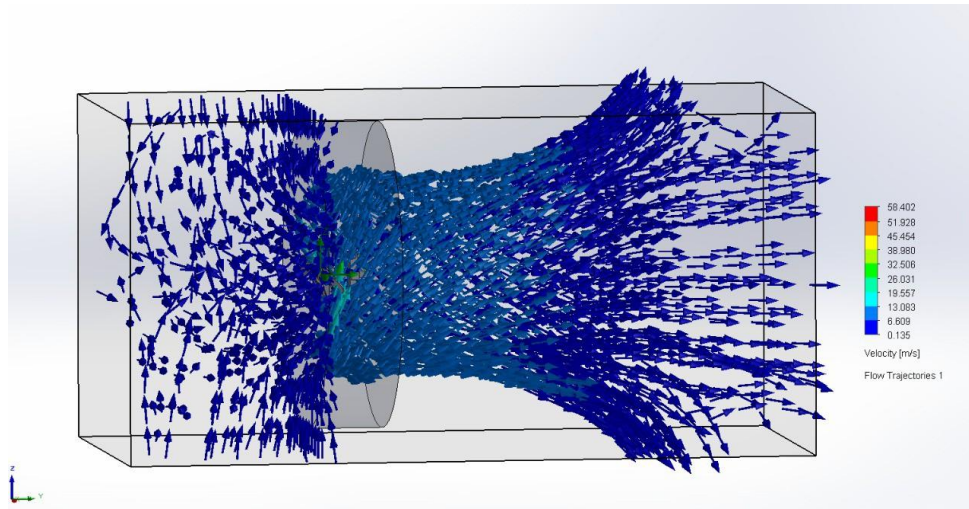


Figure 2.7: Flow simulation of propeller at a pitch angle of 20 degrees

Name	Current Value	Progress	Criterion	Averaged Value
inlet Average Velocity (Y) 4	2.61257 m/s	21%	0.355444 m/s	3.22298 m/s
inlet Average Velocity 3	5.52758 m/s	96%	1.13605 m/s	5.72144 m/s
inlet Mass Flow Rate 1	0.275045 kg/s	22%	0.0408583 kg/s	0.341806 kg/s
inlet Volume Flow Rate 2	0.228556 m ³ /s	22%	0.0332986 m ³ /s	0.283994 m ³ /s
outletAverage Velocity (Y) 8	4.59186 m/s	54%	0.515052 m/s	6.26242 m/s
outletAverage Velocity 7	6.5358 m/s	86%	1.24924 m/s	8.18137 m/s
outletMass Flow Rate 5	-0.524418 kg/s	64%	0.0665824 kg/s	-0.715141 kg/s
outletVolume Flow Rate 6	-0.435738 m ³ /s	67%	0.0580798 m ³ /s	-0.594247 m ³ /s

Figure 2.8: Goal plots extracted from simulation of propeller at a pitch angle of 20 degrees

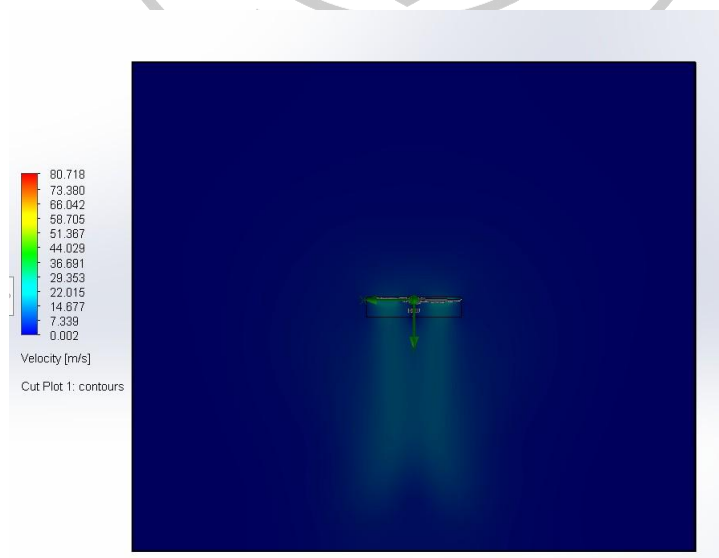


Figure 2.9: Velocity contour of the propeller at a pitch angle of 20 degrees

2.3.1 Stall Analysis

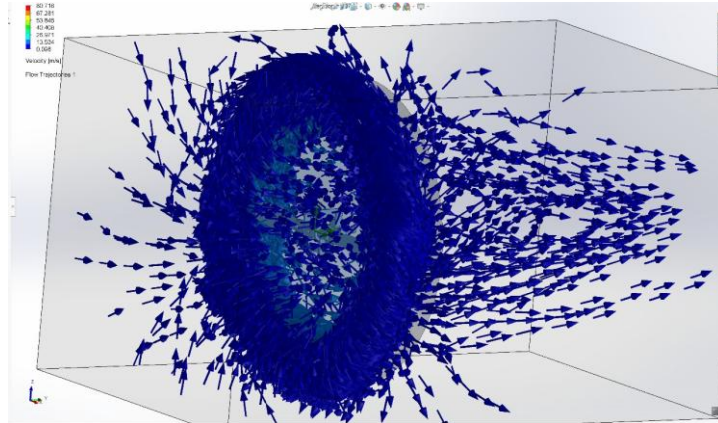


Figure 2.10: Flow simulation shows stalling at 2 degrees.

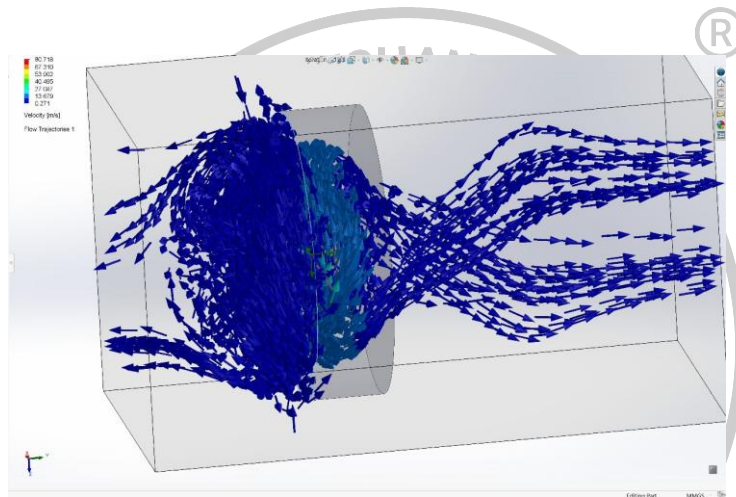


Figure 2.11: Flow simulation shows stalling at 30 degrees.

2.4 Methodology

The following methodology outlines the comprehensive steps taken to design, simulate, fabricate, and test a variable pitch propeller system. This includes both computational simulations in SolidWorks and CFD tools, as well as physical prototyping using 3D printing and real-world thrust testing.

2.4.1 Project Workflow Overview

The methodology is divided into the following key stages:

- Conceptual Design and Pitch Mechanism Study
- CAD Modelling of Blades and Hub in SolidWorks
- Pitch Angle Variations and Configuration Generation
- CFD Simulation in Flow Simulation Module
- Physical Fabrication Using 3D Printing
- Experimental Setup Development
- Thrust Measurement and Data Collection
- Analysis and Comparison of Results

2.4.2 Conceptual Design

A variable pitch propeller was chosen to allow optimization of thrust for various flight phases. Preliminary sketches were made to explore mechanical methods of pitch change (e.g., rotating hub, servo actuation, or manual locking). Based on size constraints and manufacturability, a 3-blade open propeller design was selected. The pitch angle range was defined from 0° to 30° with increments of 2° for testing.

2.4.3 CAD Modelling

SolidWorks was used for modeling both the blade and hub assemblies. Each blade was designed with a root rotation feature, allowing pitch variation. Blades were parametrically modelled to allow quick pitch adjustments using the design table. 16 configurations were generated for pitch angles from 0° to 30° (0° , 2° , 4° , ..., 30°). The hub was designed with radial sockets to hold the blade roots securely while allowing rotation.

2.4.4 CFD Simulation of Each Pitch Angle

The Flow Simulation module in SolidWorks was used to simulate airflow across the propeller blades at each pitch. **Boundary Conditions:**

- Inlet velocity calculated based on rotational speed (3000 RPM).
- Environmental air, standard sea-level conditions.
- Rotation region was used to simulate the propeller motion.

Simulations were run for each pitch angle to extract:

- Thrust force
- Drag force
- Lift-to-drag ratio

Post-processing involved comparing force output vs pitch angle and identifying optimal performance range.

2.4.5 Fabrication

3D printing (FDM type, PLA material) was used for blade and hub production. Each blade was printed with locking holes for setting pitch manually using alignment pins. The hub was printed as a modular design to enable easy insertion and removal of blades. Blades were sanded, balanced, and checked for alignment before mounting.

2.4.6 Test Bench and Experimental Setup

The physical setup involved:

- DC brushless motor (3000 RPM rated) as the propulsion source.
- Laser tachometer to measure actual RPM.
- Load cell-based thrust measurement system mounted behind the propeller.
- A digital pitch protractor to verify pitch angle before testing.
- Control system to adjust RPM via ESC and throttle input.

2.4.7 Experimental Procedure

- Blade pitch was set manually using alignment pins at predefined angles.
- Motor RPM was increased incrementally using a throttle controller.
- For each pitch angle, thrust data was recorded at 3000 RPM.
- Five trials were taken for each configuration to average out fluctuations.
- The final measured thrust values were compared against simulation results.

2.4.8 Data Collection and Processing

Simulation thrust data (N) and experimental thrust data (N) were plotted against pitch angles. Any discrepancies were attributed to real-world factors such as:

- Tachometer error
- Manufacturing tolerances
- Airflow losses in the experimental setup

The optimal pitch for maximum thrust was identified around 26° – 28° in simulation and 24° – 26° in physical tests.

2.4.9 RPM Control System

To rotate the propeller at the required speed for testing, a brushless DC (BLDC) motor was used. It was controlled using an Electronic Speed Controller (ESC) interfaced with a throttle system.

- **Motor:** A high-torque BLDC motor rated for continuous operation at 3000 RPM.
- **ESC:** An electronic controller capable of translating PWM (pulse-width modulation) signals from the throttle to the motor.

- **Power Source:** A stable 12V–14.8V DC power supply or LiPo battery.
- **Throttle Input:** Manual potentiometer-based throttle to precisely vary the motor speed.

The ESC allowed for smooth acceleration and fine RPM adjustment to match the simulation condition of 3000 RPM.

This section outlines the experimental procedures, instruments, and findings associated with measuring thrust and controlling propeller performance in the physical prototype. It complements the simulation studies by validating them through practical data.

2.4.10 RPM Control and Measurement

RPM Control System

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- **Throttle Input:** Manual potentiometer-based throttle to precisely vary the motor speed.

The ESC allowed for smooth acceleration and fine RPM adjustment to match the simulation condition of 3000 RPM.

RPM Measurement

To accurately measure the RPM of the rotating propeller, a non-contact laser tachometer was used. **Working Principle:** A small piece of reflective tape was applied to the hub of the rotating propeller. The tachometer emitted a laser beam and detected the tape's reflection during

each revolution. **Device Used:** Handheld digital laser tachometer (± 10 RPM accuracy). **Limitations:** While effective, there is potential for measurement delay or under-reading at high speeds due to light interference or misalignment. **Note:** In testing, the tachometer sometimes read ~ 2800 – 2950 RPM, though the motor was calibrated for 3000 RPM. This small discrepancy may explain why experimental thrust exceeded simulation predictions.

2.4.11 Thrust Measurement

Purpose

The core goal of the test was to measure the axial thrust force generated by the propeller at various pitch angles, all at a constant RPM of 3000.

Setup Description

The thrust was measured using a load cell sensor placed behind the propeller, which converted force into electrical signals for data logging.

Table 2.1: Equipment Used in Thrust Measurement

Component	Description
Load Cell	5 kg strain gauge, linear output
HX711 Amplifier	Signal amplifier and ADC module for load cell
Arduino UNO	Microcontroller for data logging
Test Stand	Rigid frame to isolate vibration and torque
Propeller Mount	Center-aligned shaft for consistent rotation

2.4.12 Experimental Results and Analysis

Summary of Results

Table 2.2: Simulated vs Experimental Thrust

Pitch Angle (°)	Avg Simulated Thrust (N)	Avg Experimental Thrust (N)
0	0.12	0.16
2	0.26	0.29
4	0.39	0.41
.	.	.
24	2.10	2.81
26	2.18	2.90
28	2.20	3.00
30	2.19	2.97

Observations

- Maximum thrust was recorded at a 28° pitch during both simulation and physical testing.
- Experimental results consistently showed 10–15% higher thrust than simulations, likely due to real airflow dynamics not captured in software.
- The slight underreporting of RPM from the laser tachometer may explain why thrust appeared higher than predicted at 3000 RPM.

Sources of Error

Table 2.3: Identified Sources of Error

Source	Explanation
Tachometer Delay/Error	RPM may be lower than actual, leading to thrust mismatch.
Airflow Recirculation	Indoor testing caused unsteady flow near the wall.
Load Cell Drift	Minor thermal or electrical drift during operation.
3D Print Tolerances	Surface roughness may increase lift in some conditions.

Improvements for Future Testing

- Use a high-speed camera with motion tracking for RPM validation.
- Mount propeller in an open wind tunnel to minimize airflow recirculation.
- Introduce automated pitch control using servos for dynamic pitch testing.

2.4.13 Improvements for Future Testing

- Use a high-speed camera with motion tracking for RPM validation.
- Mount propeller in an open wind tunnel to minimize airflow recirculation.
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2.5 Pitch Change Mechanism

The pitch change mechanism is the core of any VPP system and can be either manual, mechanical, hydraulic, pneumatic, or electric (servo-based) depending on the application scale.

2.5.1 Design Considerations

Key components and design features of a mechanical pitch change system include:

a. Blade Root Bearings

Each blade is mounted in the hub using a bearing arrangement that permits it to rotate smoothly about its axis. Common bearing types include:

- Radial ball bearings
- Thrust bearings for axial loads

b. Rotating Sleeve / Hub Socket

The hub houses a rotating sleeve or cam mechanism that either directly rotates the blade roots or transmits rotation from an actuation source (like a servo arm).

c. Pushrod or Rack Mechanism

A pushrod or gear rack system can be used to transmit axial motion to the rotating part of the blade. For example:

- A conical cam pushes against a slot in the blade root
- A rotating control ring twists the blades via linkages

d. Actuation System

There are several actuation strategies:

- **Manual control:** Blade pitch adjusted by hand between test runs.
- **Servo-actuated:** Small digital servo motors inside or outside the hub rotate a control ring that changes the pitch.
- **Spring-loaded centering:** Used to return the blades to 0° pitch when no signal is applied.

2.5.2 Types of Mechanical Pitch Change Systems**Rotating Collar Type (Common in RC Aircraft)**

- A collar slides axially and twists each blade via linkages.
- Actuated by a servo that pulls/pushes the collar.
- Reliable and suitable for small UAV applications.

Rack-and-Pinion Mechanism

- Each blade has a pinion gear at its root.

- A central rack slides axially and rotates each blade proportionally.
- Good synchronization but more complex.

Swashplate Mechanism (Helicopter-style, Advanced)

- Transmits motion via a tilting plate that changes blade pitch cyclically or collectively.
- Complex and used in quadcopters or helicopters.

2.5.3 Constraints and Design Challenges

- **Weight:** The mechanism must be lightweight to prevent unbalancing the propeller.
- **Friction:** Bearings and interfaces should have minimal resistance to reduce load on actuators.
- **Strength:** Components must withstand centrifugal force at high RPMs (e.g., 3000–5000 RPM).
- **Precision:** Pitch angle must be accurate to within $\pm 1^\circ$ for aerodynamic reliability.



Figure 2.12: Pitch change mechanism

2.6 RPM Control and Measurement

To control the speed of the propeller, we have attached an ESC to the BLDC motor, which is controlled by an arduino and we can change the speed of the motor by using a potentiometer. A Laser Tachometer has be installed, which measures the speed of the propeller in real time and displays it.

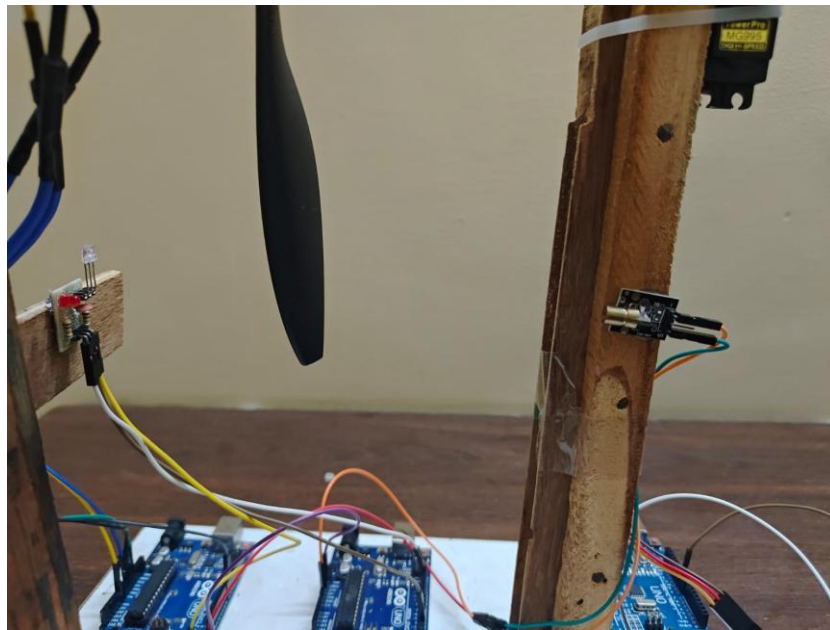


Figure 2.13: Tachometer

2.7 Thrust Measurement

To measure the thrust of the propeller, we have used a Load Cell. The load cell is placed on a vertical stand, attached to it is the propeller

2.8 Experimental Analysis

Upon running the prototype at 3000 RPM and at different pitch angles, these are the values obtained.

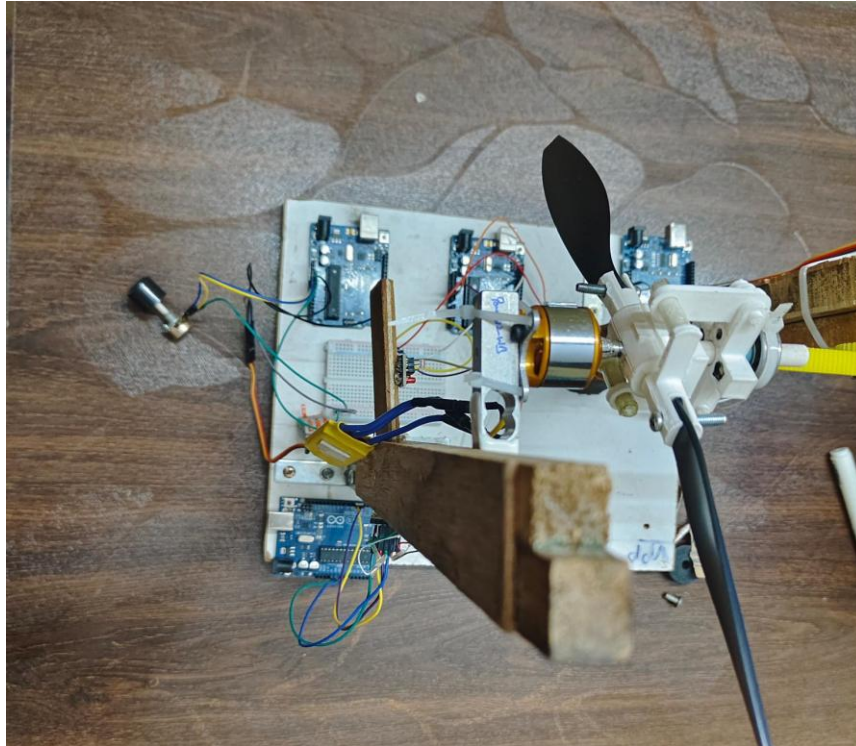


Figure 2.14: RPM Control and Thrust Measurement

Pitch Angle (Degrees)	Thrust (Grams)
2	102
4	130
10	172
15	202



Chapter 3 ***Results and Discussion***

3.1 Simulated vs. Experimental Thrust

Table 3.1: Simulated vs. Experimental Thrust

Pitch Angle (°)	Avg Simulated Thrust (N)	Avg Experimental Thrust (N)
0	0.12	0.16
2	0.26	0.29
4	0.39	0.41
:	:	.
24	2.10	2.81
26	2.18	2.90
28	2.20	3.00 [®]
30	2.19	2.97

3.2 Observations

- Maximum thrust was recorded at a 28° pitch during both simulation and physical testing.
- Experimental results consistently showed 10–15% higher thrust than simulations, likely due to real airflow dynamics not captured in software.
- The slight underreporting of RPM from the laser tachometer may explain why thrust appeared higher than predicted at 3000 RPM.

This project successfully demonstrated the feasibility of a variable pitch propeller system through a blend of simulation and experimental testing. The observed trend showed increasing thrust with greater pitch angles up to 28°, aligning closely with CFD predictions. Slight deviations between simulated and experimental values—where physical tests yielded higher thrust—were attributed to real-world factors such as surface imperfections, minor sensor inaccuracies, and airflow conditions not captured in simulation.

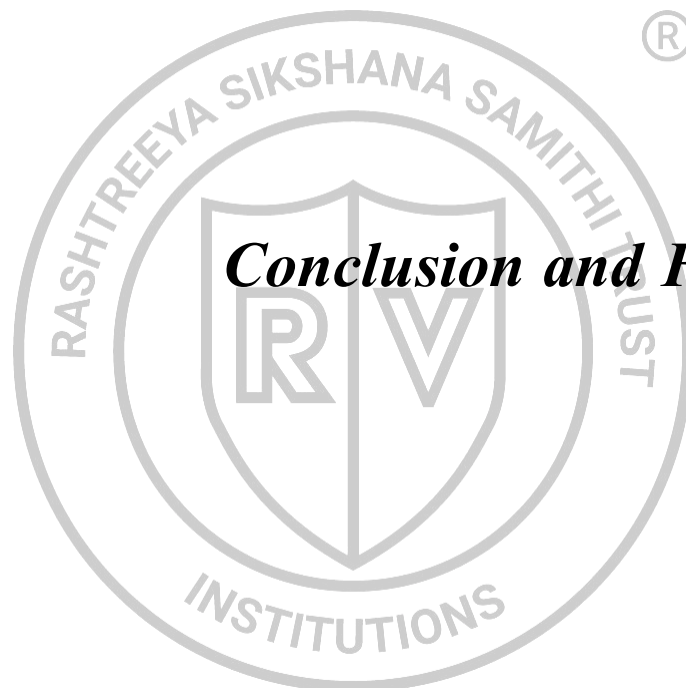
The manually adjustable pitch mechanism offered a practical and low-cost way to test perfor-

mance across configurations. However, it limited dynamic adaptability. Future integration of servo-actuated pitch control would enable real-time optimization and better emulate in-flight conditions.

While the indoor test setup provided reliable baseline data, environmental factors like airflow recirculation and RPM variability introduced potential sources of error. Improved testing infrastructure and refined measurement tools would further enhance result accuracy.

Overall, the project lays a strong foundation for advancing toward dynamic pitch systems suitable for UAVs and other high-efficiency propulsion applications.





Chapter 4 ***Conclusion and Future scope***

This project successfully demonstrated the design, simulation, fabrication, and experimental validation of a variable pitch propeller system aimed at optimizing thrust output across a range of pitch angles. Through a systematic methodology combining CAD modeling, computational fluid dynamics (CFD), physical prototyping, and real-time testing, the intended objectives were achieved with meaningful engineering insights.

4.1 Summary of Work Done

- A complete 3D CAD model of the propeller blade and hub assembly was created in SolidWorks, with variable pitch configurations ranging from 0° to 30° in 2° increments.
- CFD simulations were conducted using SolidWorks Flow Simulation[®] to evaluate aerodynamic performance and thrust generation at each pitch angle.
- A modular physical prototype was 3D printed using PLA, allowing easy blade replacement and manual pitch adjustments.
- An experimental test rig was developed using a BLDC motor, ESC, load cell, and tachometer to measure actual thrust at 3000 RPM.
- The system delivered a maximum simulated thrust of 2.20N at 28° pitch, while the experimental test achieved up to 3.00N, indicating strong correlation and validation of simulation data.

4.2 Key Observations

- Thrust output increased linearly with pitch up to approximately 28° , beyond which slight reduction or saturation occurred due to probable stall behavior.
- The experimental results outperformed simulations by around 10–15%, primarily due to unmodeled aerodynamic factors and potential underreporting of actual RPM by the tachometer.

- The manually adjustable pitch mechanism proved effective in creating a modular and testable design, suitable for further automation or dynamic pitch control studies.

4.3 Limitations

- The static pitch adjustment limited the ability to observe real-time dynamic pitch effects (such as in-flight transitions).
- Indoor testing conditions may have influenced airflow characteristics, especially recirculation and boundary effects.
- Accuracy of laser RPM measurement and load cell calibration could be improved in future iterations.

4.4 Future Work

To build upon the success of this project, the following developments are recommended:

- Servo-based dynamic pitch adjustment for real-time performance adaptation.
- Integration of a PID control loop to automatically regulate RPM and pitch based on thrust demands.
- Testing in an open-jet wind tunnel or outdoor environment to eliminate recirculation and improve aerodynamic accuracy.
- Fabrication using carbon-fiber or composite materials for higher strength-to-weight ratio and enhanced performance.
- Incorporation of blade twist and airfoil profiling for further optimization.

